ECSE-6320 Advanced Computer Systems
Project 1 Report

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Abstract

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1 Introduction

The goal of this project is to analyze the performance of SIMD Instructions and how their performance is affected by the datasize, size of the vector, memory alignment, data size, and compiler optimization.

To do this I will run multiple experiments across three different kernels:

1. Vector Multiply and Accumulate: $\vec{y} = a \times \vec{x} + \vec{y}$
2. Vector Elementwise Multiplication: $z_i = x_i * y_i$
3. 3-Element 1D Stencil: $y_i = a \times x_{i-1} + b \times x_i + c \times x_{i+1}$

For each kernel I will adjust the size of the vector using different vectoriziation instructions (SSE, AVX, AVX512). Adjust memory alignment, element size (FP32 vs FP64), number of elements provided, and compiler flags for autovectorization.

Performance will be evaluated using the Linux `timespec` to measure execution time of the vector execution while isolating overhead from I/O operations, and verification.

2 Pre-Experiment Hypotheses

Pre-Experiment Hypothesis

Hypothesis:

As the size of the vector grows the amount of time required for computation will decrease, however with diminishing returns.

Reasoning:

Larger vector width means that we can process more data in a single instruction. However, more data may have to be loaded from memory on larger vectors causing more pipeline stalls and latency as data comes from outside the CPU cache.

Predicted quantitative outcome:

Table 1: Predicted SIMD speedup relative to scalar baseline

Vector ISA (bits)	Vector Size Increase	FP32 Speedup	FP64 Speedup
SSE2 (128)	4× (FP32) / 2× (FP64)	∼3.3×	∼1.8×
AVX2 (256)	8× (FP32) / 4× (FP64)	∼6.4×	∼3.2×
AVX-512 (512)	16× (FP32) / 8× (FP64)	∼12×	∼5.6×

Pre-Experiment Hypothesis

Hypothesis:

Shifting the memory alignment off of the 4 Byte or 8 Byte boundary will significantly affect the performance of the SIMD instructions. The instructions will likely still run, however they will not run efficiently and may even be slower than the scalar counterpart.

Reasoning:

CPUs are most efficient when the data is alligned to the memory boundary since CPUs will fetch memory multiple bytes at a time. If the memory is shifted off the memory alignment boundary it will have to do multiple data fetches to fill the buffers. This will increase the number of pipeline stalls and reduce instruction throuhput.
Predicted quantitative outcome:

Table 2: Predicted SIMD Slowdown relative to SIMD with 0 byte offset

Offset Bytes	**FP32 Slowdown**	**FP64 Slowdown**
1	∼0.5×	∼0.25×
2	∼0.5×	∼0.25×
3	∼0.5×	∼0.25×
4	∼1×	∼0.25×
5	∼0.5×	∼0.25×
6	∼0.5×	∼0.25×
7	∼0.5×	∼0.25×

3 Correctness Validation

3.1 Validation Method

3.2 Validation Results

Table 3: Correctness validation summary

Check	Expected	Observed

4 Measurement Discipline

4.1 Experimental Setup

- **Hardware:** CPU: x86_64 AMD Ryzen AI 7 350 w/ Radeon 860M (SIMD Support: SSE, SSE2, AVX, AVX2, AVX-512), Memory: 32GB LPDDR5 @ 5600MT/s
- **OS / Kernel:** Pop_OS! 24.04 Running Linux 7.0.11-76070011-generic
- **Compiler & Flags:** GCC 16.2 compiling with -O2 so that I can control autovectorization.
- **Isolation techniques:** Ran on CPU0 with 100% affinity, and -20 niceness (ensure no context switching)
- **Trials per condition:** 5 trials per condition
- **Timer / Instrumentation:** clock_gettime(CLOCK_MONOTONIC)

4.2 Statistical Treatment

5 Results

5.1 Summary Figure

Figure 1: Caption describing the primary result.

5.2 Secondary Figures

(a) (b)

Figure 2: Side-by-side comparison of …

5.3 Results Table

Table 4: Quantitative results

Configuration	Mean	Std. Dev.	N

5.4 Relevant Code

```c
int main(void) {
    // ...
    return 0;
}
```

Listing 1: Benchmark kernel (C)

```cpp
#include <chrono>

int main() {
    auto start = std::chrono::high_resolution_clock::now();
    // ...
}
```

Listing 2: Measurement harness (C++)

```rust
fn main() {
    let start = std::time::Instant::now();
    // ...
}
```

Listing 3: Benchmark harness (Rust)

6 Negative / Failure Case

Negative / Failure Case

What failed:

Evidence:

Root cause (if known):

7 Honest Anomalies

Honest Anomalies

Anomaly 1:

Anomaly 2:

8 Discussion

9 Conclusion

A Additional Data / Raw Logs